

Denver opens November 1. Three products are on track, and the fourth needs a decision this week.

Lucas Diverdi | AI.205: AI Integration in Business I | Prof. Jeff Eyt | Homework #5

Prepared for: Marcus (CEO) and Jennifer (CFO) Portfolio:

<https://github.com/Ldiverdi13/greenthreads-operations-analyst>

THE DECISION WE NEED THIS WEEK

Jennifer, we are asking for three approvals before Friday, September 11. **Sign the EcoFleece Hoodie order (\$32,300)**, since its deadline falls on Saturday. **Sign the Active Shorts order (1,200 units, \$26,400)** to ship by ocean. **Move \$29,300 from the catalog-SKU budget line to the focus-SKU line**, because the line paying for the launch is short by exactly that amount. We are not requesting any contingency money; the \$7,000 already committed to expedite stays in reserve as the Joggers' backup.

Marcus, the short answer is yes. Denver opens November 1 with all four products ordered. Three are projected on the floor for opening day (I). The shorts follow one to two weeks later at best (I), and Section 3 explains why that beats paying to rush them.

Every figure carries a tag. (V) means verified against the source files. (I) means inferred, a planning estimate and not a forecast. (U) means it cannot be verified from anything we hold. All datasets were pulled July 14, 2026; documents are current as of their own dates.

1. Where Denver stands, 52 days out

Operations had one goal this quarter: get the four launch products onto the Denver sales floor on time. Doors open Sunday, November 1 (V). The space starts accepting goods October 13, and everything has to be in the store by October 19 so the team has ten business days to receive, inspect, and set the floor (V, client rulings). Measured against that date, three products are on track and one is not.

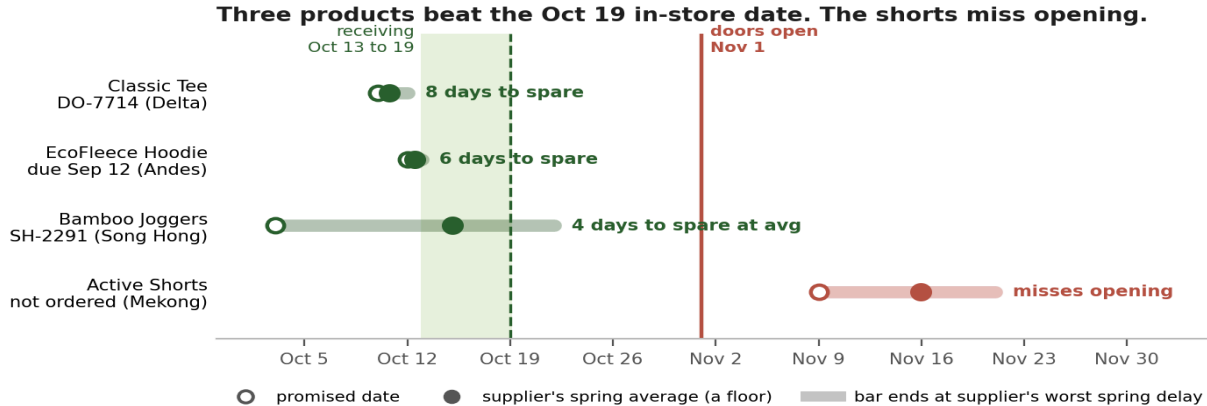


Figure 1. Projected arrival by product (I). Promised dates for the Hoodie and Shorts assume an order placed by Sept 12 and today, using catalog lead times. Supplier history covers spring orders only, so for autumn every dot and bar end is a floor rather than a worst case.

The Classic Tee (PO DO-7714) is promised October 10, and Delta Organics has averaged under a day late, which puts it around October 11 with eight days to spare (I). The Bamboo Joggers (PO SH-2291) are promised October 3. Song Hong has been late on all 12 of its completed orders by an average of 12.1 days (V), which lands the Joggers around October 15 with four days of slack (I). Twelve for twelve is a real pattern and a thin sample at the same time. Every one of those orders was also promised between January and July, so 12.1 days is the least we should expect this autumn rather than the likely outcome. At Song Hong's worst spring delay the Joggers arrive October 22, three days past the in-store date and still ten days before opening (I).

The EcoFleece Hoodie has not been ordered. Its order-by date is Saturday, September 12, so in practice the order goes out Friday (V). Placed on time, it lands around October 12 with six days to spare (I).

The Active Shorts are the problem. Their August 13 order-by date passed without a purchase order, and the client confirmed the order was missed rather than deferred (V). A 1,200-unit order placed today arrives around

November 9 on the catalog's 60-day lead time, or November 16 at Mekong's spring average (I). Both dates are best cases. Mekong has never delivered on its promised date in our data, and the catalog does not say whether its lead time includes ocean freight and customs (U). Ocean freight cannot get the shorts to the opening. They carry \$43,200 of gross margin, about a quarter of the launch (V).

The chart also shows a smaller catch. The Tee and the Hoodie are projected to arrive before October 13, when the space can first accept goods (I). The rear alley dock is appointment only and shared with two other tenants (V), so Operations should book delivery appointments from October 13 onward and give both suppliers that date now.

2. How we got here: one quarter, one lesson

Every piece of this quarter pointed at the same idea, which is that a promised date is not a plan. In HW1 our team mapped the Operations function and ranked two AI opportunities at the top, flagging high-risk shipments and scoring supplier reliability. Both made the list for the same reason. Time is the scarcest thing this launch has, and both tools catch a delay early enough to act on it.

In HW2 we used AI to read the documents behind the Joggers order. Song Hong's confirmation letter says it does not anticipate any issues meeting October 3. The master supply agreement between the same two companies calls that date a good-faith estimate, excuses nearly every cause of delay, and limits our only remedy to cancelling before the goods ship (V, §4 to §6). The same review found that no document stated the date Denver actually needed the goods, which is the first input our own receiving procedure, OPS-04, asks for.

In HW3 we moved from documents to numbers. Both Vietnam suppliers were late on every completed order, and four departments turned out to be planning against four different dates, so the recommendation was to confirm the real in-store date before anything else. In HW4 I built the tool that does this work going forward, the GreenThreads Operations AI Analyst. It is a Claude project loaded with 25 source files and rules that make it label every claim and refuse to invent numbers. Its testing paid off right away. On August 27 it reported that no purchase order for the Active Shorts existed anywhere in the files, two weeks after the order-by date had passed (V, test log). On September 8 the client confirmed the order had been missed (V). That finding shapes the decision below.

3. The decision

The Active Shorts are the only real choice on the table, so the options below are for them. Each is priced in the same terms: how much of the launch's \$177,800 in gross margin is physically on the floor when doors open, what extra money it takes, and what sales are at risk while the shorts are missing. Margin is shown before freight and duty, because no freight figure exists in any file (V).

Option	What happens	Launch margin on the floor Nov 1	Extra spend	What we give up
A. Ocean only (recommended)	Order all 1,200 today by ocean. Shorts arrive one to two weeks after opening at best (I).	\$134,600 (76%) (I)	None beyond regular freight	Roughly \$6,000 to \$11,000 of shorts sales during an 8 to 15 day gap (I)
B. Ocean plus air on the first 300	Order all 1,200 today, ask Mekong to finish 300 early, and fly those 300.	Up to \$145,400 (82%) if they land in time (U)	Up to the full \$7,000 committed to expedite, with no quote in hand (U)	The Joggers' only backup money, to recover part of a gap the shorts' sales will largely make up later
C. Air-freight everything	Order all 1,200 and fly the full order.	Up to \$177,800 (100%) if it lands in time (U)	Unknown until quoted; anything over \$7,000 draws on the remaining \$13,000 of contingency	Could spend most of the reserve on one product with no quote in hand

We recommend Option A. Waiting does not lose the \$43,200 in the shorts, because those units still sell once they arrive. What waiting costs is sales during the gap. At Austin's pace, the shorts' share of daily gross margin

is about \$745 (I), so an 8 to 15 day gap puts roughly \$6,000 to \$11,000 at risk (I). That figure can move in either direction. It assumes no shopper buys a Tee or Hoodie instead, which overstates it, and it uses an established store's pace, which may understate opening week (U). Against that, flying just 300 units could consume the full \$7,000 committed to expedite, and no quote exists yet (U).

That \$7,000 does more work protecting the Joggers. They can make the opening, and with the shorts on the water they will be the only bottoms on the floor opening week (V, catalog categories). At Song Hong's worst spring delay they miss the in-store date by three days (I), and autumn could be worse. Waiting on the floor also does not mean waiting on the order. All 1,200 shorts get ordered now, because every day the order sits moves arrival a day later (I), and the client ruled that the four-product assortment is not being reopened (V). We will still ask Mekong for a production date and get one air quote, since neither commits any money (V). If a quote comes in well below the sales at risk in the gap and the Joggers are confirmed on a vessel, flying part of the order is worth a second look. Otherwise the shorts come by ocean.

Two things have to happen before the Mekong order goes out. The first is funding. The four launch products cost exactly \$110,000, while the budget line meant to pay for them holds \$80,700 (V). The missing \$29,300 sits, to the dollar, in a separate line for the four catalog-only products the client said not to plan or budget for Denver (V). The Tee and Joggers orders already committed come to \$51,300, so the Hoodie order alone pushes the line \$2,900 past its budget (V). Under purchasing policy FIN-02 that needs Jennifer's approval regardless of size, on top of the approval any invoice over \$5,000 already requires (V). Both lines belong to Operations in the budget (V), so this is a move inside our own lines that still needs her signature. Three payments totaling \$11,009.77 already sit on the catalog line, including \$3,889.28 to Song Hong, which makes none of the catalog-only products (V). Finance will need to decide whether those get recoded as part of the move.

The second is a control gap. Finance shows a pending \$16,668.82 payment to Mekong coded to the launch line (TXN-9118), and no purchase order sits behind it (V). It also does not match the 1,200-unit order at \$26,400 (V). Operations and Finance should reconcile it before the new order goes out so GreenThreads does not pay for the same shorts twice.

One cost still has no home. None of the budget's twenty lines covers freight (V), and regular inbound freight does not qualify for contingency, which only covers a response to a delay (V). It needs its own funding decision from Jennifer, and nobody can size it until a quote comes in (U).

Who does what. Jennifer approves the Hoodie order, the Shorts order, and the line move, plus any future expedite draw, in writing and naming the product and invoice. Operations places both orders once approved, reconciles TXN-9118 with Finance, requests Mekong's production date and one air quote, tracks Song Hong's milestones weekly, and books dock appointments. The Store Manager owns the receiving window. The assistant calculates and flags; a person makes every call.

4. The contradiction we were part of

For most of the quarter, Operations and HR were both planning for a store that opened October 12, and the lease said that could not happen. Each side had good data. Our catalog's order-by dates all land on October 12 once each lead time is added (V). HR's offer letters start associates September 28, and the handbook requires two weeks of paid training before opening, which also lands on October 12 (V). The lease delivers the space on or about September 1 and says build-out customarily takes six to eight weeks, which made October 13 the earliest possible opening (V). No document ever set October 12 as a date. It was a number the order-by math produced, and then everyone planned to it.

We did not split the difference. We escalated it through our PM as a conflict for the client to rule on, and the client set the dates we use now: goods accepted from October 13, required in-store October 19, doors open November 1 (V). October 12 is retired.

What remains open sits between us and HR. October 19 is now both the day merchandise is due and the day two weeks of paid training has to begin (V). Receiving runs October 13 through 19, and OPS-04 requires every carton inspected within 24 hours of arrival (V), which comes to about 198 cartons for the three products arriving in that window (V, catalog arithmetic). To be more certain the window works, we need HR to confirm who will be on payroll and on site before October 19. Staffing that week is HR's plan to make; the question is ours to raise.

5. Where AI helped, where we corrected it, and where a person stays in charge

AI made this quarter faster where it mattered. It pulled every delivery and remedy term out of the supply contract in minutes, held four documents in view at once to find where they disagreed, and recomputed supplier statistics from 96 purchase orders on demand. The assistant passed all nine of its tests on grounding. Every number it produced was recomputed against the source files and matched, and it refused all four attempts to make it invent something, including a request for a freight cost that does not exist (V, test log).

It also got things wrong, and those failures are the most useful part of the record. In HW2, a plain “summarize the contract” prompt produced penalty and on-time clauses the agreement does not contain, because the model wrote what supply contracts usually say instead of what this one says. Requiring a quoted passage for every term fixed it. That same first pass skipped sections 5 and 7 in the middle of the contract, which held the finding that mattered most. In HW3 I lost points for options that were not priced in the same unit and for claims stated more confidently than the evidence allowed, which is why every figure here carries a tag. In HW4 break-testing, the assistant would not stay short. A Monday-morning shipment question got over a thousand words that opened with methodology. I added a rule to lead with the action and keep answers brief, and prioritization improved immediately while length dropped only about 25% (V, test log). Grounding rules held nine times out of nine; the length rule only moved the needle. The reliable fix is stating a length in the request, which worked every time I tested it.

The most useful correction came last. My first draft of this brief, written with the assistant, recommended holding an air-freight option on 300 shorts, released if the quote fit inside the \$7,000. That test checked whether the money was available, not whether flying the shorts paid for itself. A review question exposed it: why not open with three products and let the fourth follow? Pricing the gap in the same unit as the freight reversed the recommendation. It is the clearest case this quarter of a person catching what the AI missed, and it is why a human stays on every decision that spends money.

Testing also caught a flaw in our own earlier work. Song Hong’s 12.1-day average only counts orders that arrived. Three Song Hong orders were still open in the data, and if they had not landed by late August, two were already later than Song Hong’s worst completed order (V, test log). The data cannot tell us whether they landed, so the true average is likely higher (I). The assistant has one more limit that no rule fixes: it only knows what its context file tells it. In August it correctly warned that the in-store date existed nowhere, and after the client’s September 8 rulings that warning was wrong until a person updated the file. Someone at GreenThreads has to own that file.

A person stays in the loop on anything that releases money, changes a supplier relationship, or goes in front of the CFO. No order is placed, cancelled, or expedited on an AI flag alone. The tags only work if people pass them along, because an inference repeated without its label becomes a fact by the second retelling.

6. What we assumed and could not verify

Limit	Why it matters	Where the answer comes from
No dataset contains a Denver store (V)	Every Denver figure is inferred from other stores, never observed	Denver POS data once the store trades
Supplier history is spring only; no autumn history exists (V)	Averages are floors for an autumn launch, not expectations	Cannot be recovered; the client confirmed it was never captured
Song Hong’s average rests on 12 orders (V)	Real pattern, thin sample; treat as directional	Weekly SH-2291 milestones going forward
No freight, duty, or expedite cost in any file (V)	All margins are gross of freight; Options B and C cannot be priced yet, and baseline freight has no budget line	Forwarder quotes, requested now; quotes commit nothing
Whether catalog lead times include ocean freight and customs (U)	If they do not, the shorts land later than Nov 9	Mekong, at order placement

Limit	Why it matters	Where the answer comes from
How many shoppers buy another product when shorts are missing (U)	Moves the \$6,000 to \$11,000 gap estimate in either direction	Denver sales data after opening
Build-out completion Oct 20 is a point estimate inside a 6 to 8 week range (V)	At the top of the range, finishes and signage slip; the opening is at risk, receiving is not	Contractor progress reports
Case brief says 3,200 sq ft; lease says about 2,400 RSF (V)	Stockroom capacity for about 198 cartons is unconfirmed	Store Manager, on site
Three of four buys are not whole cartons (V)	The $\pm 2\%$ tolerance gives the Joggers an accept band of 931 to 969 units	Packing lists requested on every PO

What this work shows

The tools did the reading and the arithmetic. The judgment was knowing which date was never real, which number was a floor, and which question belonged to another department. That is what I would bring to any team adopting AI: build the assistant, prove where it breaks, and only then trust it near a decision.

Tools, method and verification. Document analysis and the HW4 assistant ran on Claude (Opus, reasoning tier) in a Claude project; HW3 also used ChatGPT's reasoning tier. Every figure in this brief was recomputed in Python with pandas from the source files and matches the formulas in the HW3 workbook linked in the portfolio. This brief was drafted with AI assistance, and each number was checked against the files before it went on the page.

Sources. GreenThreads case brief and client data room rulings (rounds one and two); GT_SKU_Catalog, GT_Ops_Inbound_Shipments, GT_Finance_Denver_Budget, GT_Finance_Spend_Transactions, GT_Finance_Austin_Store_Daily; Song Hong PO confirmation SH-2291 and master supply agreement; OPS-04 v2.1; FIN-02 v1.4; Denver lease (July 10, 2026); HR offer letter and handbook §3. MIT Sloan Management Review, "Making generative AI work in the enterprise"; Storytelling with Data blog; Mollick, "Thinking like an AI" (2025).